Exercise 8 - SQL foundations

1 SELECT product name,
price,
CASE
WHEN PRICE > 1000 THEN 'Expensive'
WHEN PRICE BETWEEN 100 AND 1000 THEN Mid range
WHEN Price < 100 THEN Budget
END AS Price category
FROM products;

Products.

product name	price	Price category
laptop	1200.00	Expensive
phone	800.00	Mid-range
keyboard	45.00	Budget
Monitor	800.00	mid range
Mouse	25.00	Budget

2 SELECT ~~pts~~ customer name,
amount,
CASE
WHEN Amount ≥ 1000 THEN 'high value'
WHEN AMOUNT BETWEEN 500 AND 999 Then 'Medium value'
WHEN AMOUNT < 500 THEN 'LOW VALUE'
END AS order value category FROM orders;

customer name	amount	order value
Alice	150.00	low value
Bob	560.00	medium value
Charlie	9,99.99	Medium value
Diana	45.50	low value
Ethan	1200.00	high value

→ orders

3 SELECT employees name,
department,
Salary,
CASE
WHEN department = 'IT' AND Salary > 70000 Then 'Senior IT'
WHEN department = 'HR' AND Salary > 55000 Then 'Experienced HR'
ELSE 'Staff'
END AS Position Level
From employees;

employees

emp name	department	Salary	Pos level
John	IT	75 000	Senior IT
Sara	HR	60 000	Experienced HR
Mark	IT	75 000	Staff
Lucy	Finance	95 000	Staff
Tom	HR	55 000	Staff

4 SELECT student name,
score,
CASE
WHEN Score ≥ 90 THEN 'A'
WHEN Score BETWEEN 80 AND 89 THEN 'B'
WHEN Score BETWEEN 70 AND 79 THEN 'C'
WHEN Score BETWEEN 60 AND 69 THEN 'D'
WHEN Score < 60 THEN 'F'
END AS grade
From students;

Students

Student name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5 SELECT delivery id,
delivery time minutes,
CASE
WHEN delivery time minutes ≤ 30 THEN 'Fast'
WHEN delivery time minutes Between 31 AND 60 THEN 'on Time'
WHEN delivery time minutes > 60 THEN 'Late'
END AS performance
FROM deliveries;

delivery ID	DTM	performance	
1	45	on time	→ deliveries
2	80	Late	
3	30	Fast	
4	65	Late	
5	100	Late.	

6 SELECT issue types,
priority,
CASE
WHEN priority = 3 THEN 'HIGH
WHEN priority = 2 THEN 'Medium'
WHEN priority = 1 THEN 'Low'

END AS priority level
FROM tickets ;

tickets

issue type	prority	priority label
login issue	1	low
slowdown	3	high
slow system	2	medium
email error	2	medium
password reset	1	low

7 SELECT id, Student ID,
CASE
WHEN % ≥ 90 THEN Excellent
WHEN Between 75 AND 89 THEN 'Good'
WHEN <75 THEN 'needs improvement'
END AS attendance status
FROM attendance;

attendance

Student ID	Att percentage	Att status
1	90	excellent
2	80	needs improvement
3	96	excellent
4	50	needs improvement
5	100	excellent

8 SELECT product_id,
stock_qty,
CASE
WHEN stock_qty = 0 THEN 'out of stock'
WHEN stock_qty BETWEEN 1 AND 5 THEN 'low stock'
WHEN stock_qty > 5 THEN 'in stock'
END AS stock_status
FROM products_inventory;

inventory.

product_ID	stock_qty	stock_status
1	5	low stock
2	0	out of stock
3	25	in stock
4	10	in stock
5	3	low stock

9 SELECT subjects,
enrolled_students,
CASE
WHEN enrolled_students ≥ 25 THEN 'Large'
WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
WHEN enrolled_students < 10 THEN 'Small'
END AS class_size_category
FROM classes;

subject	enrolled_stu	class_size_cat
maths	30	Large
english	25	Large
science	15	Medium
art	5	Small
history	20	Medium

→ classes

10 SELECT payment id,
payment method,
amount,
CASE
When payment method = 'cash' AND amount ≥ 200 THEN
'Eligible for Discount'
Else 'Not Eligible'
END AS discount eligibility
FROM payments;

payments

ID	payment method	Amount	discount eligibility
1	Card	50.00	not eligible.
2	cash	200.00	eligible for discount
3	card	150.00	Not eligible
4	paypal	75.00	Not eligible
5	cash	300.00	eligible for discount.